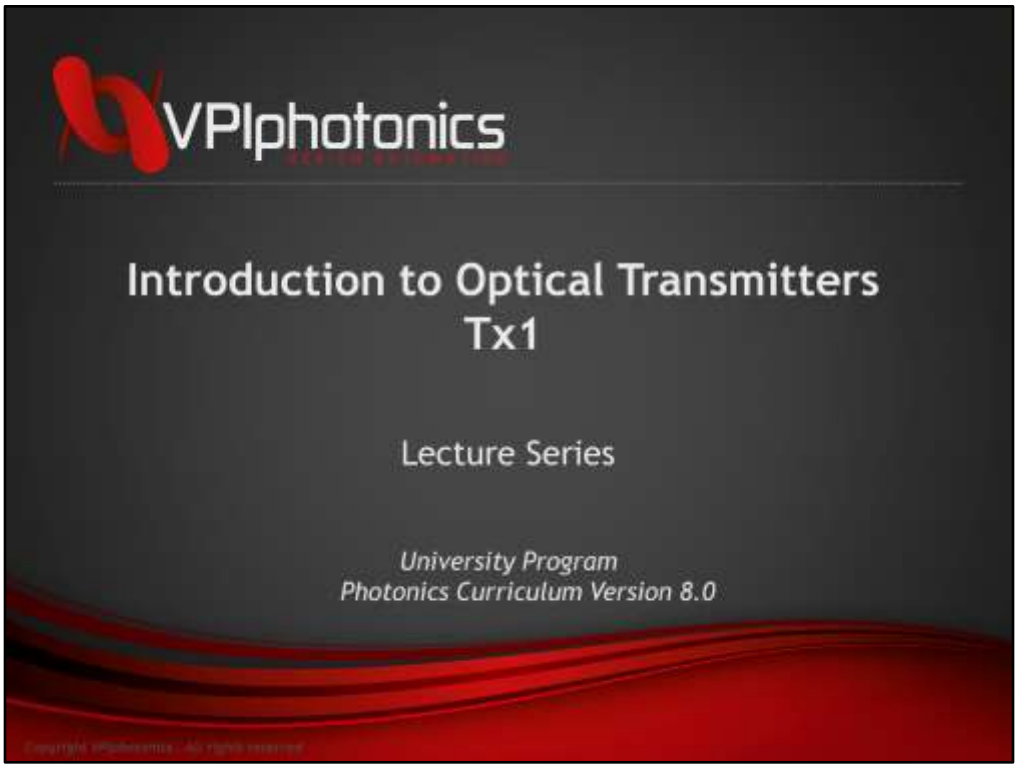
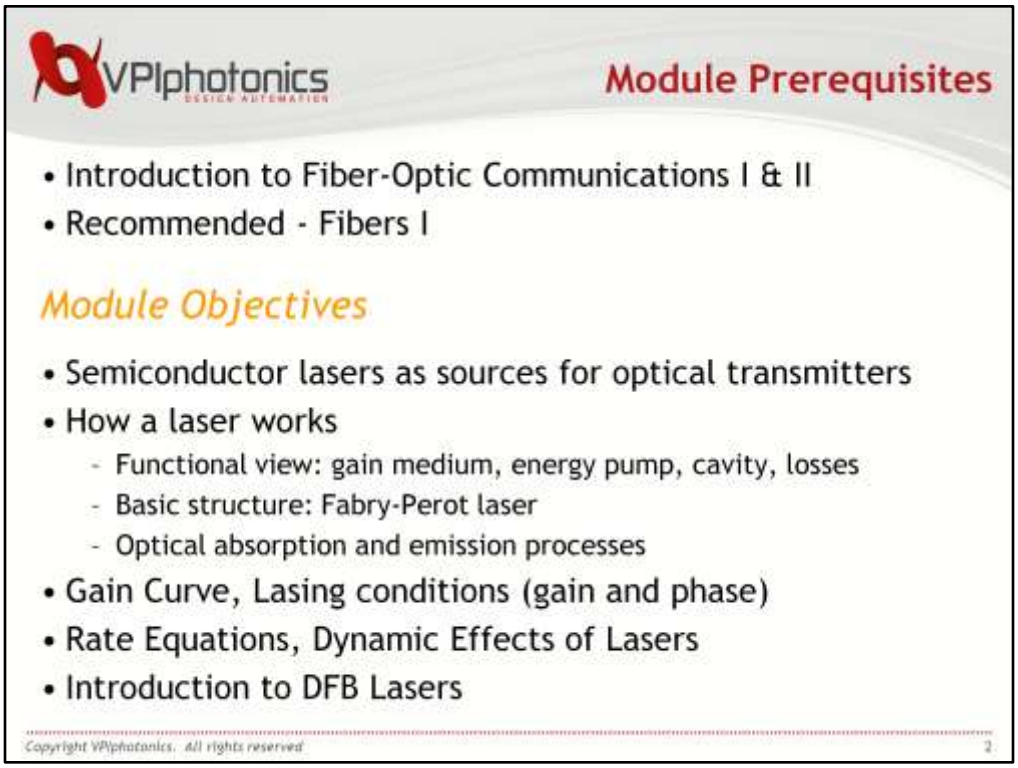
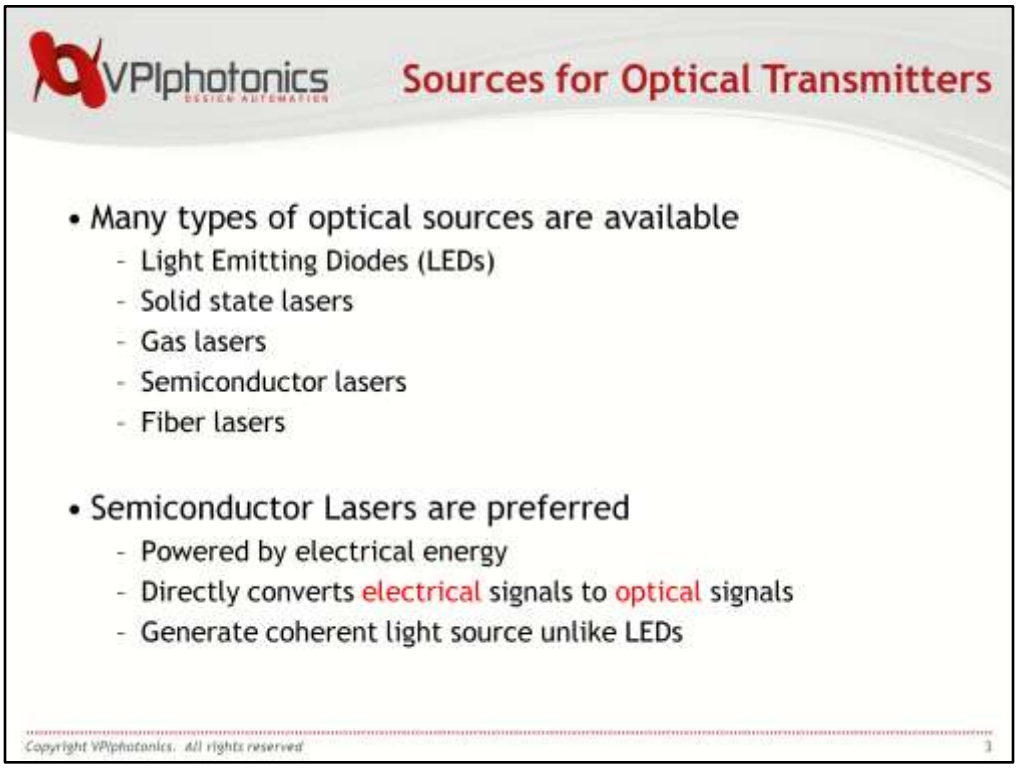

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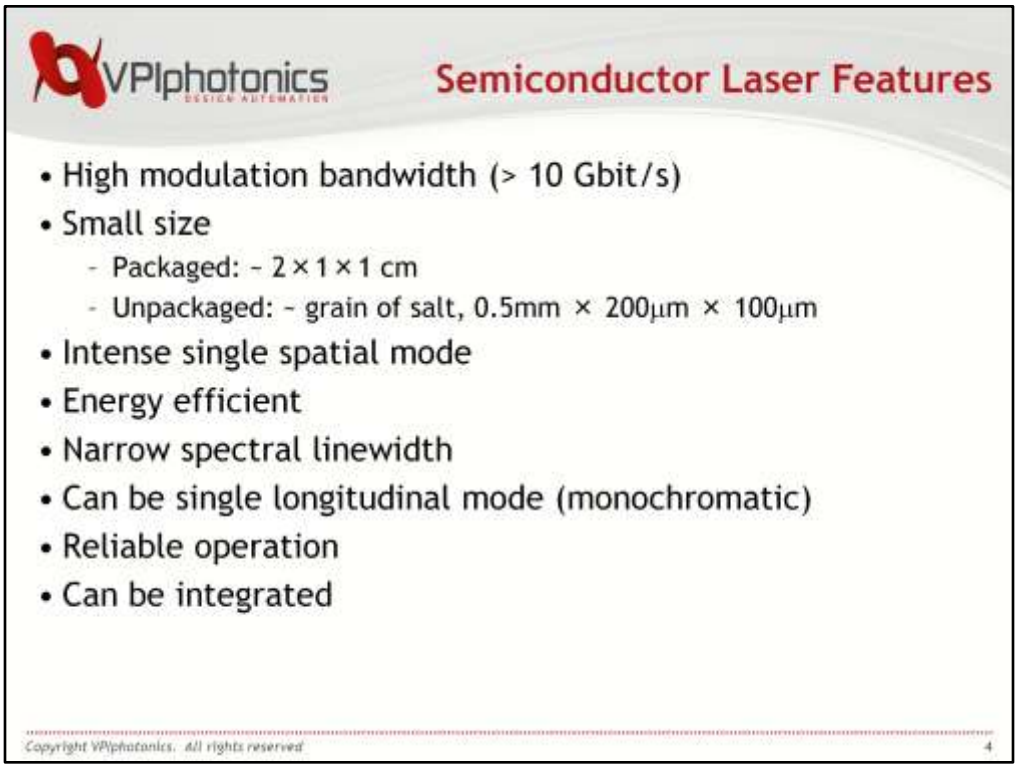
Module prerequisites and objectives – edit to suit the specific requirements of your course.



Many types of lasers can be used as the optical source for optical transmitters.

For practical applications in fiber-optic communications, we need a source capable of taking in electrical input signals and converting it to optical signals (as foreshadowed in modules 1 & 2, Introduction to Fiber-Optic Communications). In this list provided above, only semiconductor lasers offer the advantages of being powered by electrical energy and the ability to directly convert electrical signals to optical signals.

Other desirable features of semiconductor lasers are shown in the next slide.



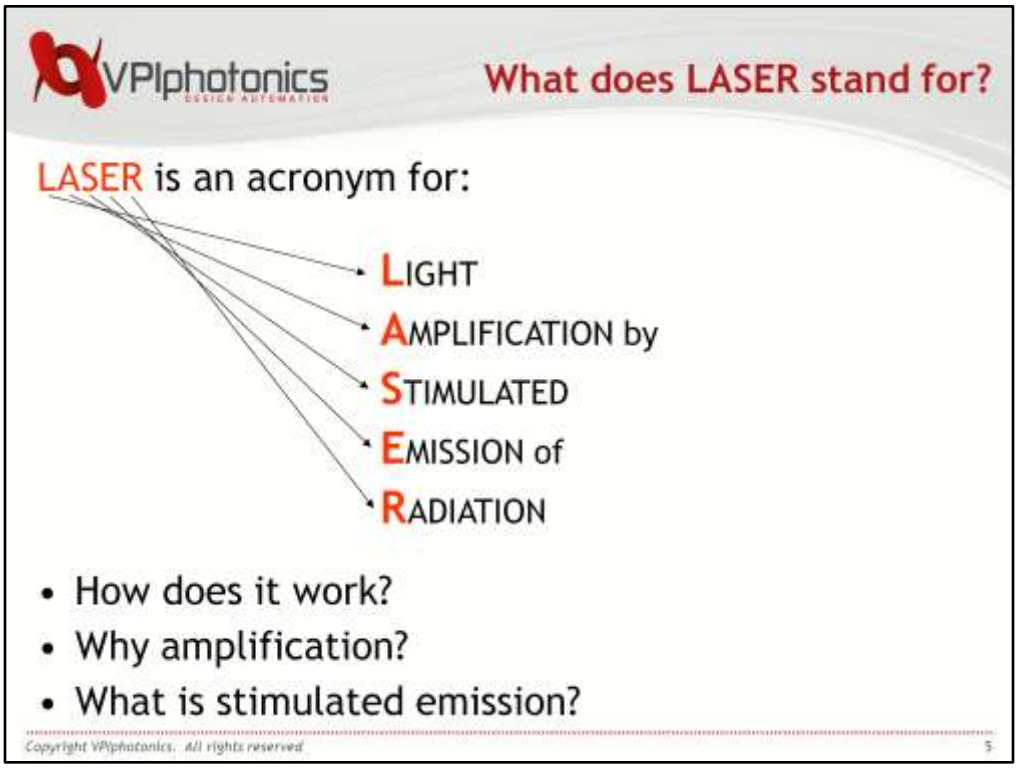
- Other laser systems (gas, solid state etc) are not capable of the high modulation speeds or gain-bandwidths that semiconductor lasers offer. For that matter, semiconductor lasers are also very compact and small (the other laser systems are large, bulky and impractical for deployment in the field). Also, the other laser systems require a powerful energy source (some solid state lasers require a high voltage and current sources in the range of thousands of volts and hundreds of amperes!) as well, whereas semiconductor lasers require modest voltage and current sources.

- Intense single spatial mode optical beam for coupling to single mode optical fibers

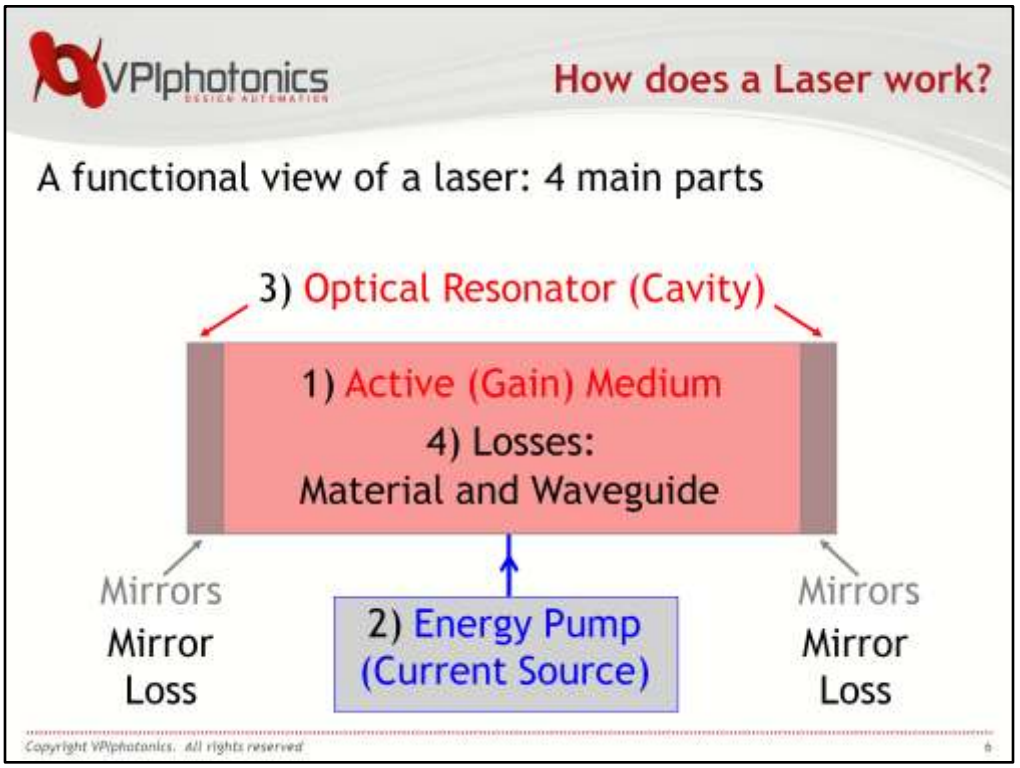
Narrow linewidth for dense WDM systems, where many closely spaced wavelength channels are required

Single longitudinal mode, along with narrow linewidth, also helps minimise fiber dispersion problems.

Semiconductor Lasers can be integrated with other active and passive optical components to realised complex photonic integrate circuits and chips.

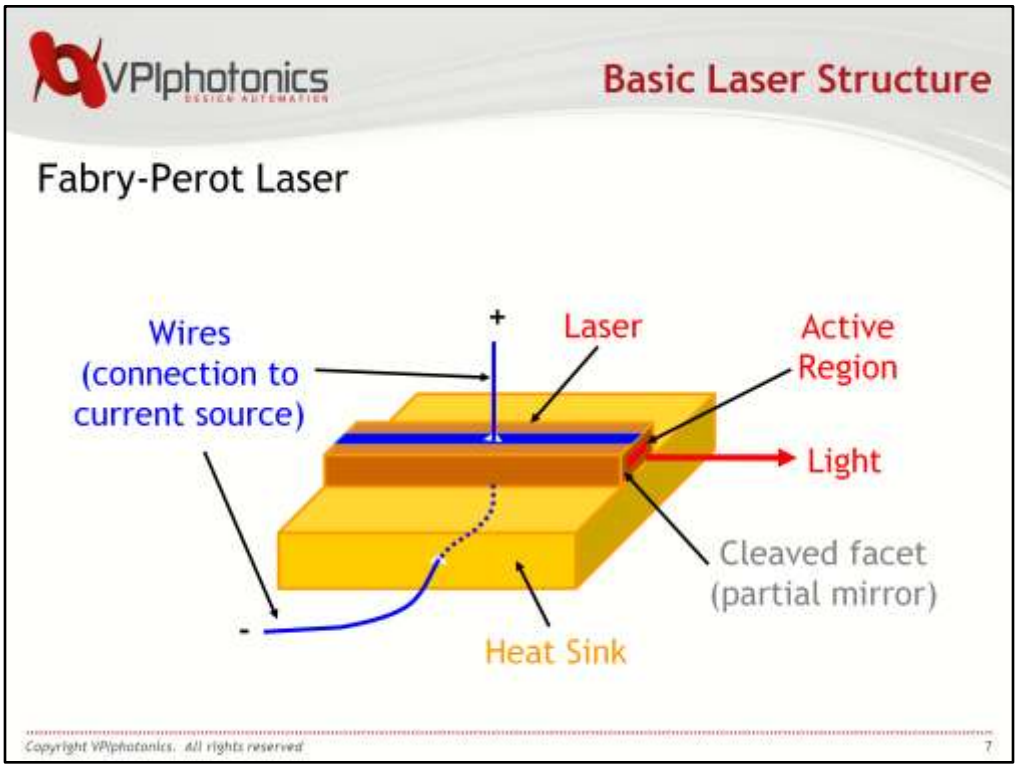


LASER is an acronym for - Light - Amplification by - Stimulated - Emission of - Radiation. In the ensuing slides, we will look at how a laser works, what is meant by “light amplification” and “stimulated emission”.



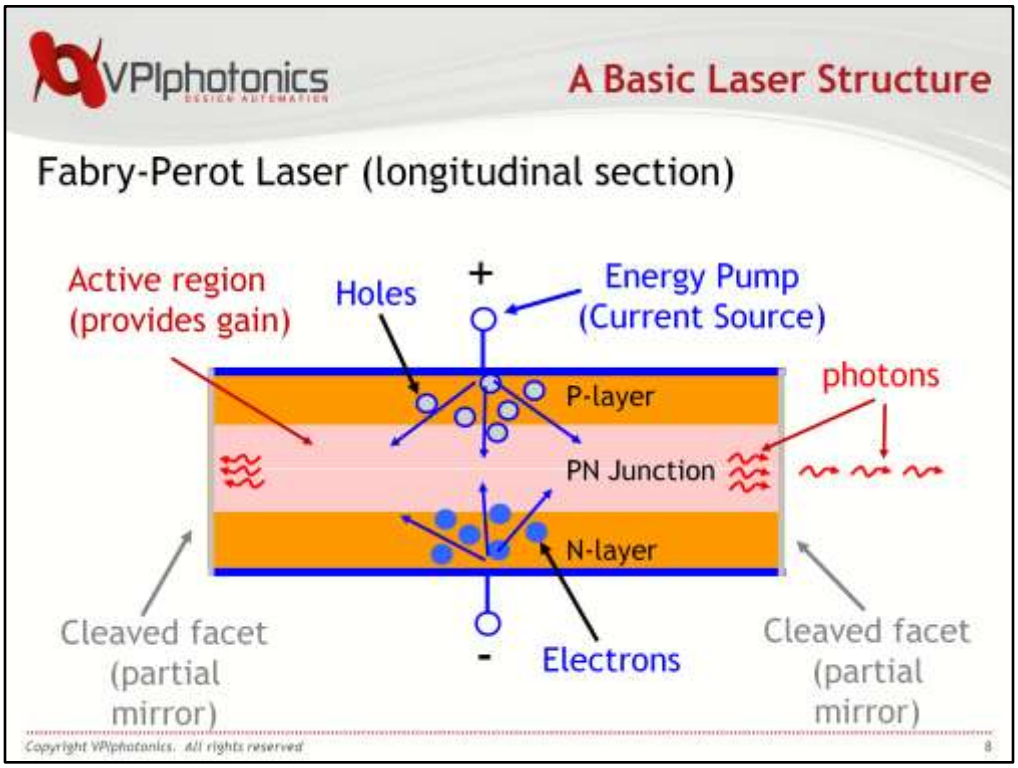
When viewed from a functional perspective, a semiconductor laser can be subdivided into four main parts:

1) An active medium that provides amplification (gain) for the light that is generated in the device. This is material that, when "excited", is capable of producing light via two optical processes (spontaneous and stimulated emission of light) that will be explained in detail later.
2) An external energy source (termed a "pump" as we are "pumping" energy into the device). For semiconductor lasers, this is a current source.
3) An optical resonator or "cavity". This is a structure that is used to provided feedback into the the active medium. Literally, it is a pair of mirrors that reflect light backwards and forwards in the active medium. This helps sustain the lasing action of the device. The cavity also define and help the laser "remember" its frequency. As shall be explained in later slides, it also governs other important characteristics of the laser. Additionally, the cavity acts as a "waveguide" to guide the light generated in the laser in the appropriate direction to maximize its amplification and the output of the laser.
4) Losses. The energy being pumped into the systems has to go somewhere; it cannot just accumulate indefinitely in the device. Apart from material losses (arising from the absorption of the generated light by the materials used to construct the device and subsequent conversation to heat energy), there are also waveguide losses (due to light lost from not coupling into the guided modes on the laser cavity). Furthermore, any light transmitted through the mirrors also counts as losses. This is important! If the mirror are lossless, then the device would not produce any light output. The losses sustained via the transmission of light from the cavity to the surrounding environment provide the required rate of dissipation of energy required to balance the energy flow into the device.



A simple working laser structure is like the one introduced in the previous slide, comprising of an active medium between two cleaved facets that act like partial mirrors. In this slide, a laser chip is shown mounted on a heat sink. This is to dissipate the heat generated in the laser.

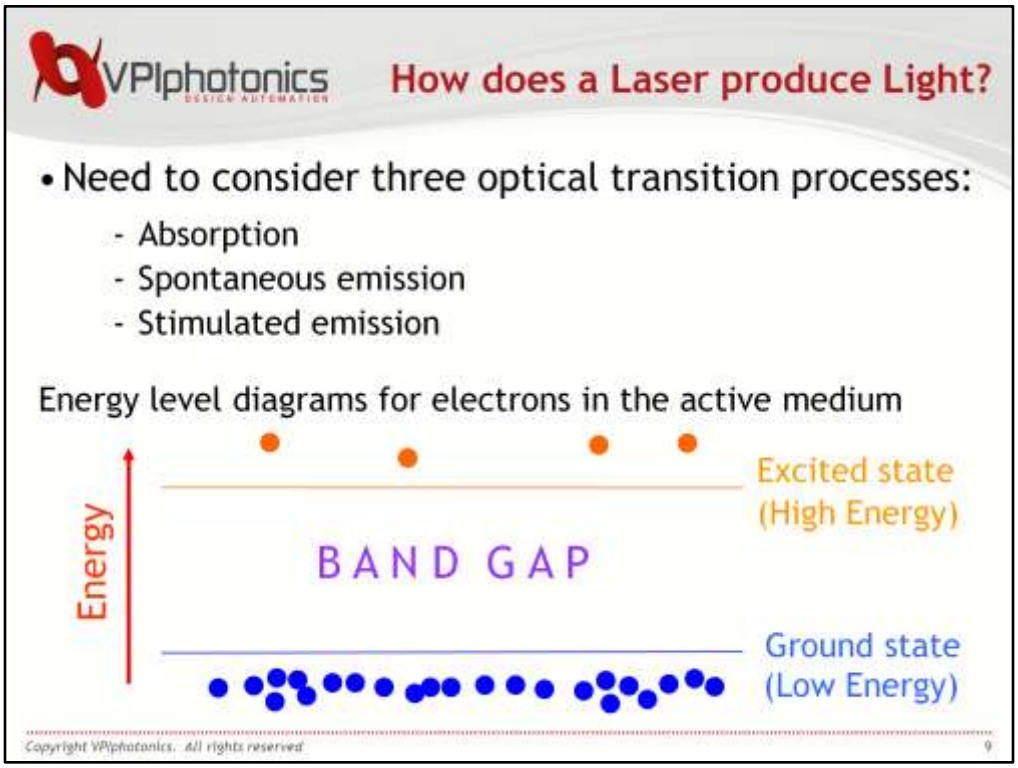
Although not shown in the diagram, it is also usual for laser chips to be mounted on temperature regulation/control devices (e.g. Thermo-electric/“Peltier-effect” cooler).



This slide shows a "simple" explanation of how a laser works:
Energy in the form of current (electrons and holes) are injected into the laser. Electrons are injected into the N-layer via the negative electrical terminal, while holes are injected into the P-layer via the positive electrical terminal. The electrons and holes combine in the active region, and the annihilation of electron-hole pairs produces photons.

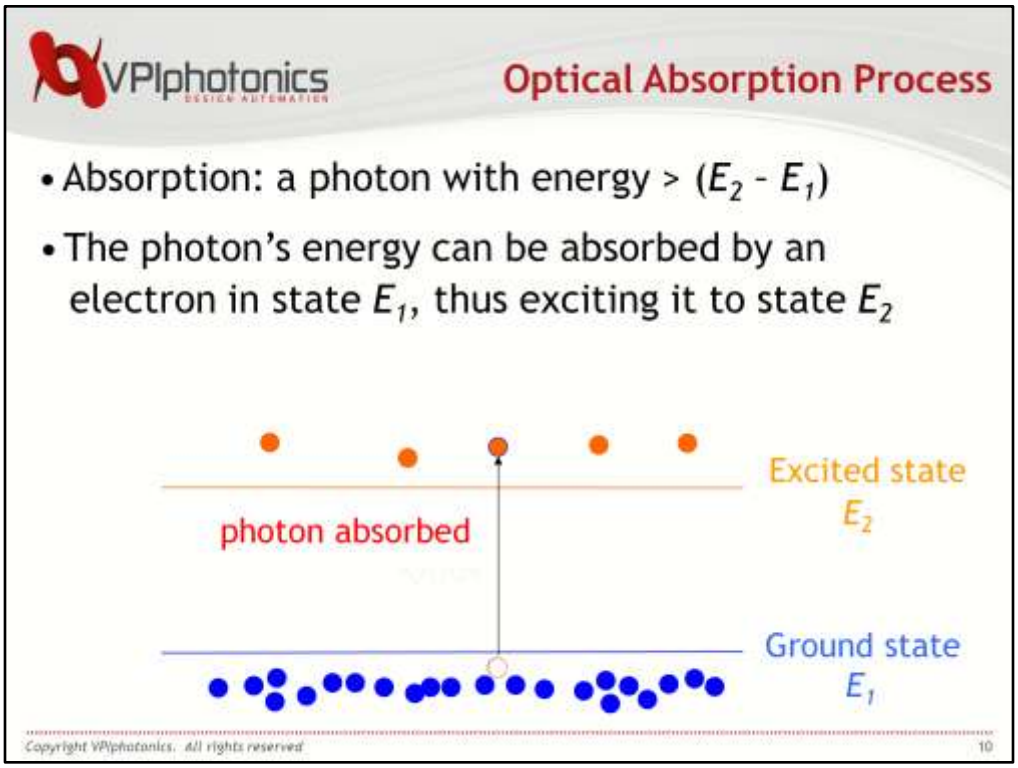
Some of these photons are lost because:
•They are not coupled into a guided mode of the waveguide structure in the laser cavity (see the module on Fibers for a discussion of guided modes) – they hit the non-reflecting/opaque sides of the laser cavity and are absorbed.
•They may be absorbed by impurities in the laser material and dissipated as heat

A portion of these photons are coupled into a guided mode of the laser's waveguide structure (which is in the direction perpendicular to the two cleaved facets of the laser). Because the cleaved facets are partially reflecting (partial mirrors), some of these photons are reflected back into the laser cavity, while the rest are transmitted and become the output of the laser. As these "guided mode" photons travel through the active medium in the laser cavity, they can stimulate the emission of additional photons. How is this possible? (see next slide…)



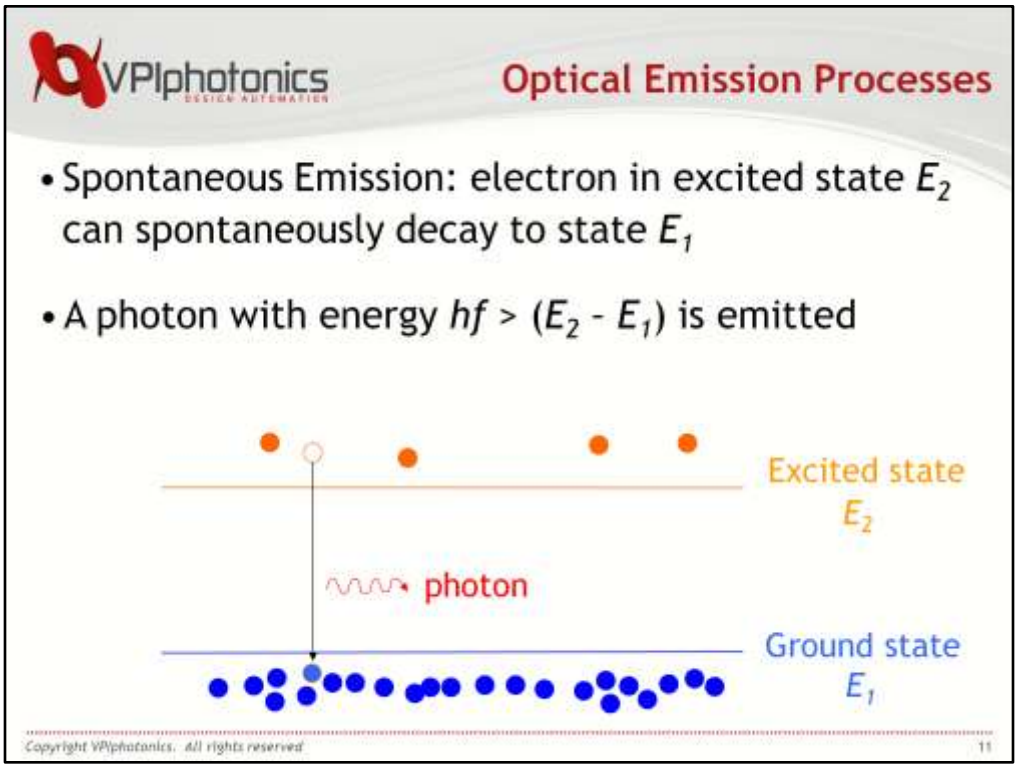
The simple explanation in the previous slides help us understand, in a high-level big-picture sense, how a laser works (produces light).

To obtain a greater, in-depth understanding of how a laser produces light, we need to consider three fundamental optical transition processes:

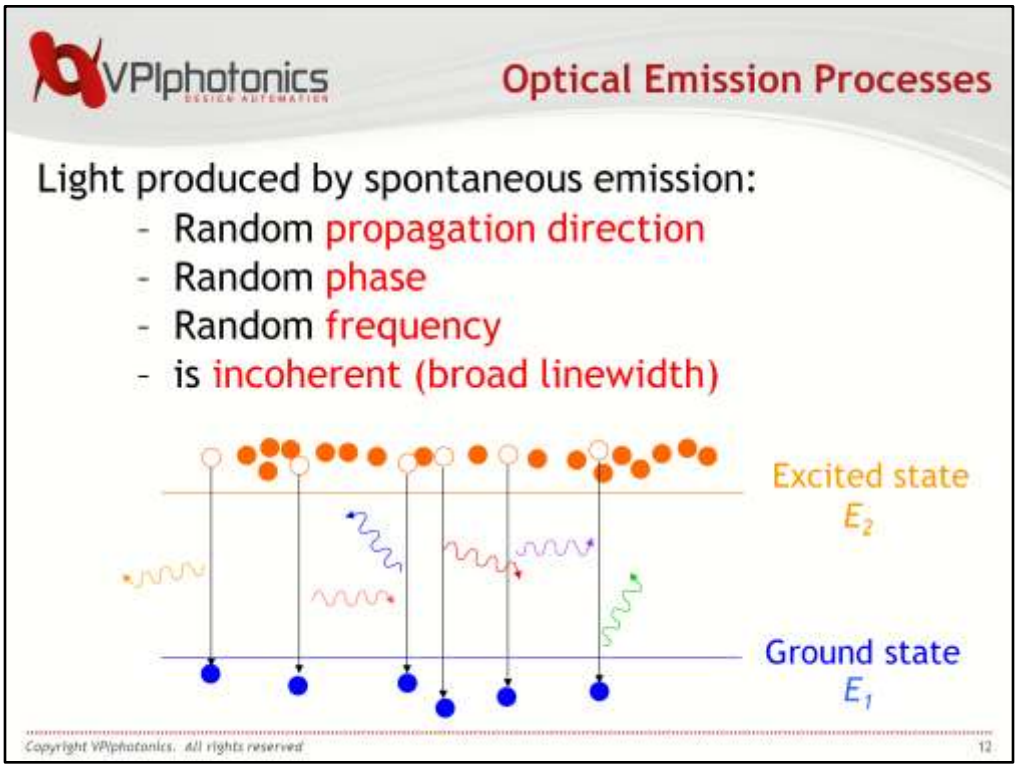


In the optical absorption process, a photon with energy $hf > (E_2 - E_1)$, which is the energy difference between the Excited state and the Ground state (i.e. the energy or band gap), can be

$(E_2 - E_1)$ to have a chance of being absorbed – as long as its energy is $hf > (E_2 - E_1)$, there will be a probability that the photon will be absorbed by one of the electrons in the ground state.



In the optical spontaneous emission, an electron in the excited state E_2



Light produced by spontaneous emission:

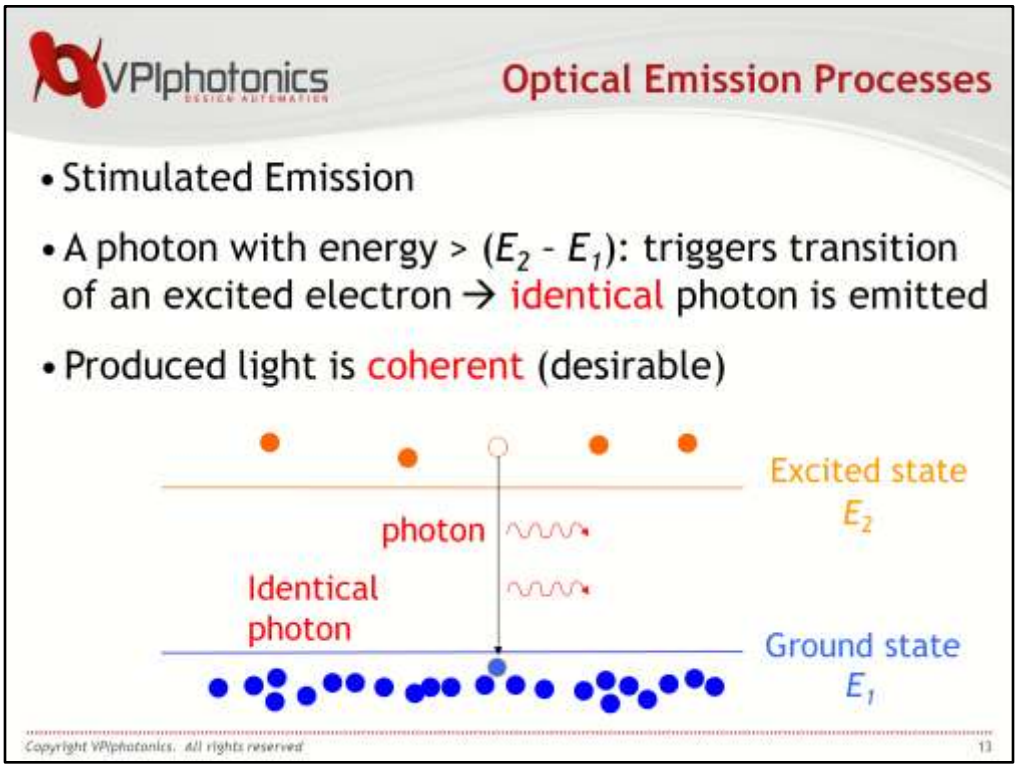
propagation direction

phase

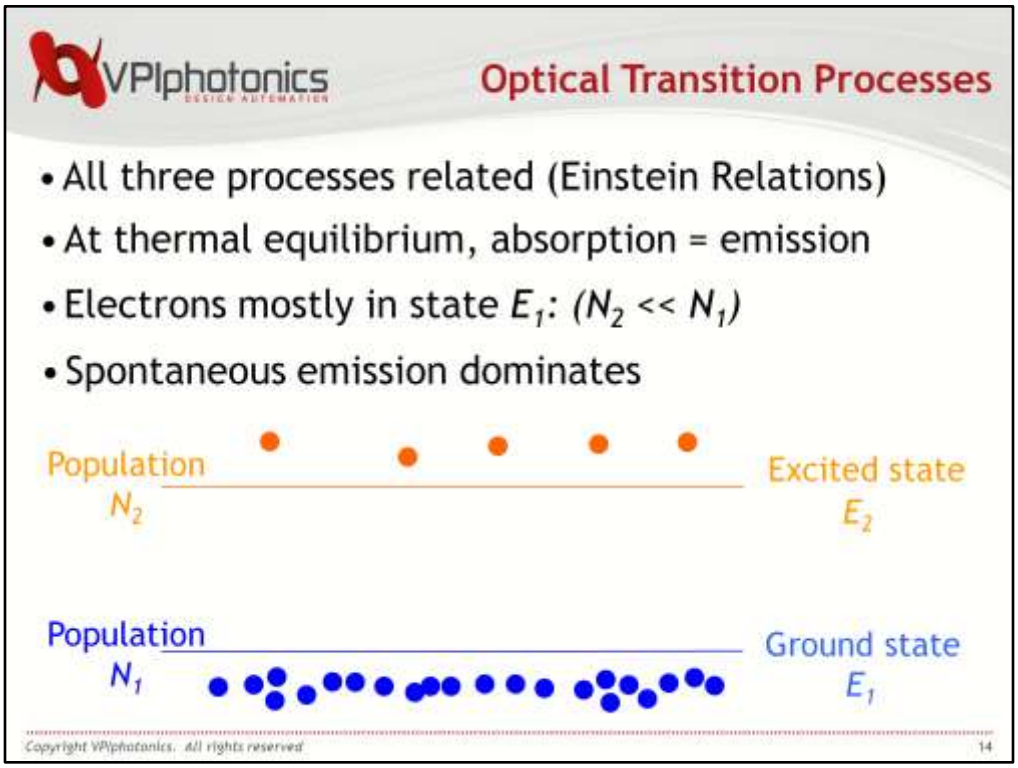
frequency

incoherent (the spectrum of spontaneous emission has a broad linewidth)

From the properties of spontaneous emission, it is clear that it is a source of noise (phase and intensity) in the laser.

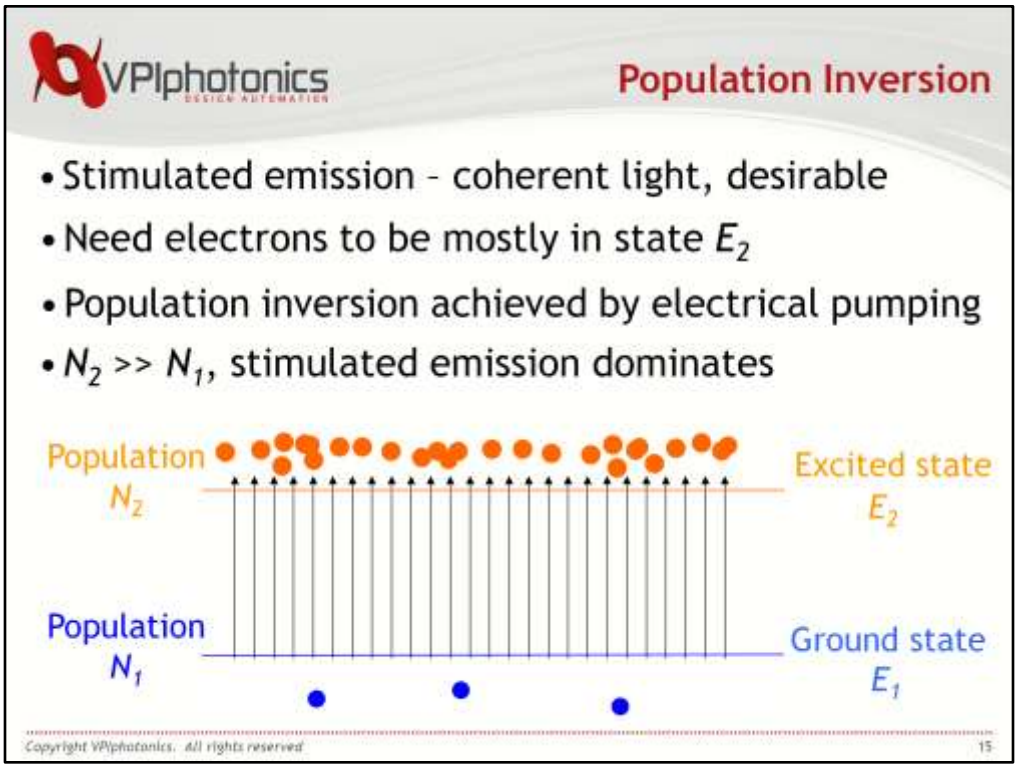


In the optical stimulated emission process, a photon with energy
transition of an electron in the excited state E_2



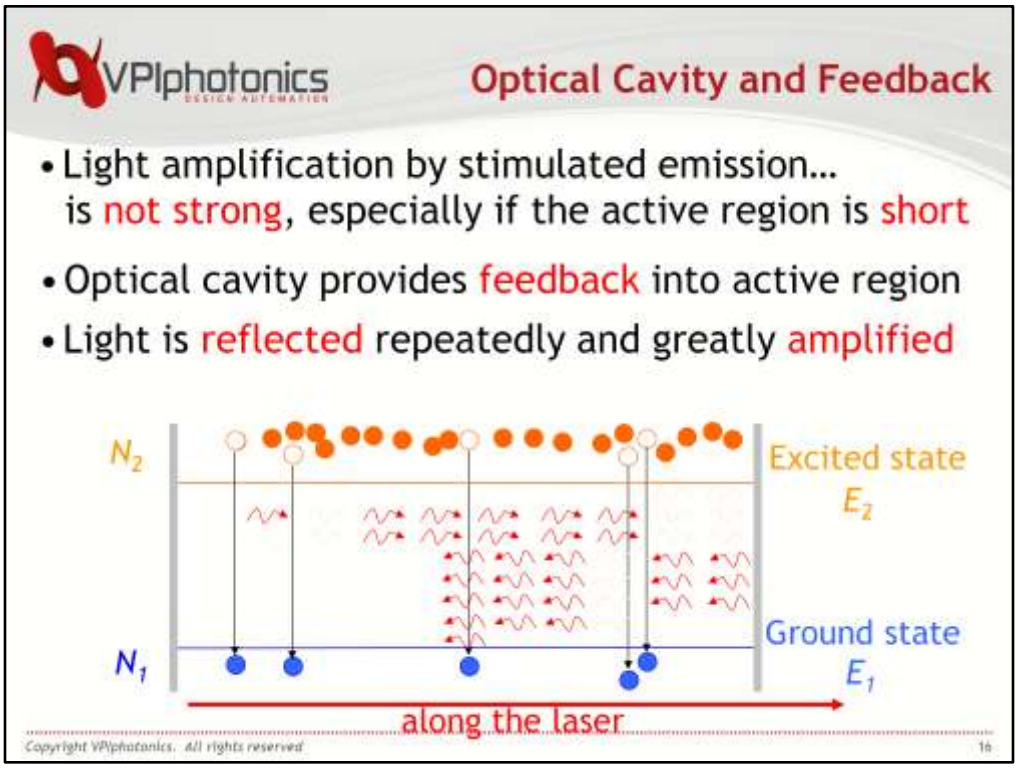
All three processes (absorption, spontaneous and stimulated emission) are related by the Einstein Relations (this should be expanded upon in later lectures/modules – for now, could be set as a reading exercise for students). These relations (and the Einstein coefficients) can be used to deduce certain properties of the light produced by a laser under specific conditions.

In particular, when the laser is at thermal equilibrium with the environment, the rate of optical absorption is equal to the rate of emission. Coupled together with the fact that electrons in the laser material is mostly in the ground state, the Einstein Relations can be used to show that the light generated by the laser under such conditions are dominated by spontaneous emission.



However, it is stimulate emission that is desirable, not spontaneous emission. In order to increase the proportion of stimulated emission, we need more electrons to be in the excited state, so as to increase the probability of stimulated emission (this is also deduced from the Einstein Relations but note that in this situation, the material is not in thermal equilibrium).

The condition where there are more electrons in the excited state than in the ground state is known as “population inversion”.

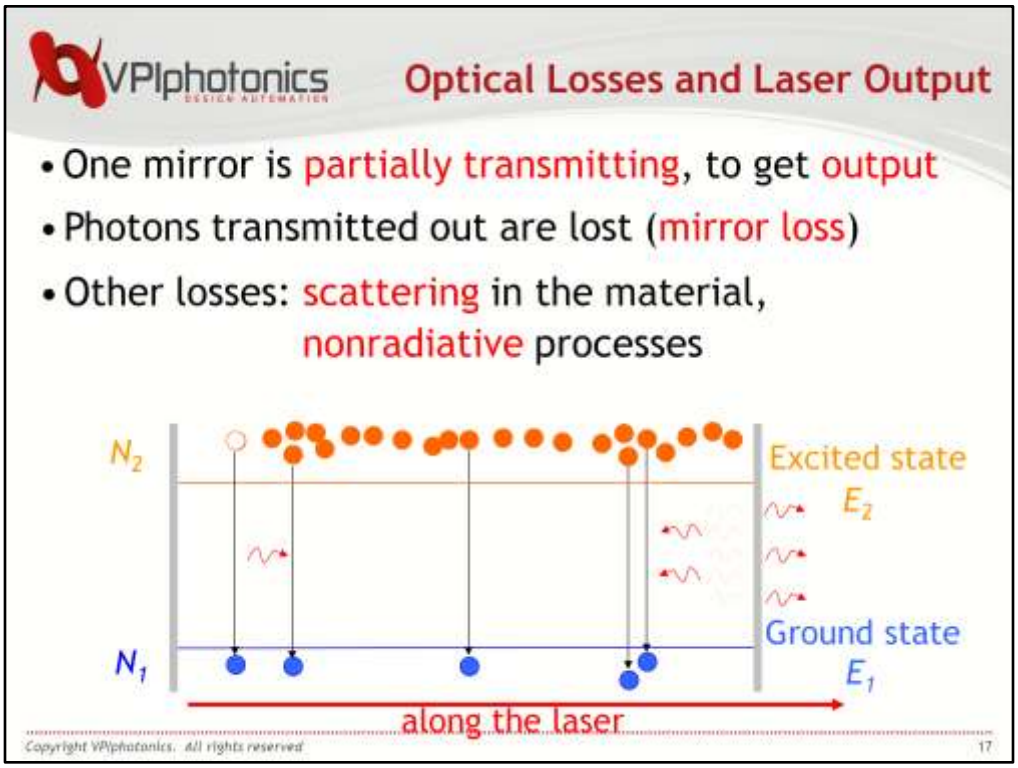


Optical feedback, resonator/cavity and laser oscillation

The amplification (gain) afforded by population inversion/stimulated emission is not strong, especially if the active region is short (as is typical in miniaturized semiconductor lasers).

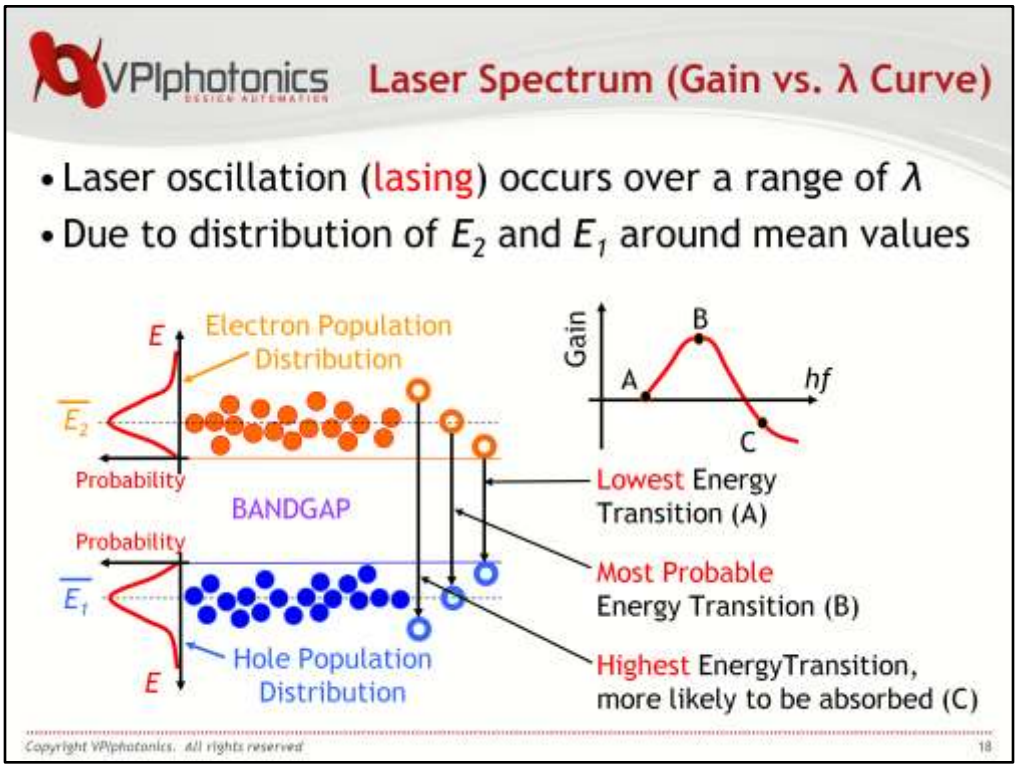
As mentioned in earlier slides, mirrors of the optical cavity or resonator reflects light backwards and forwards through the active region

Thus, stimulated emission is increased due to multiple passes through gain medium, with the reflected light amplified each time it passes through the active region. Over many reflections, the total gain would be high.

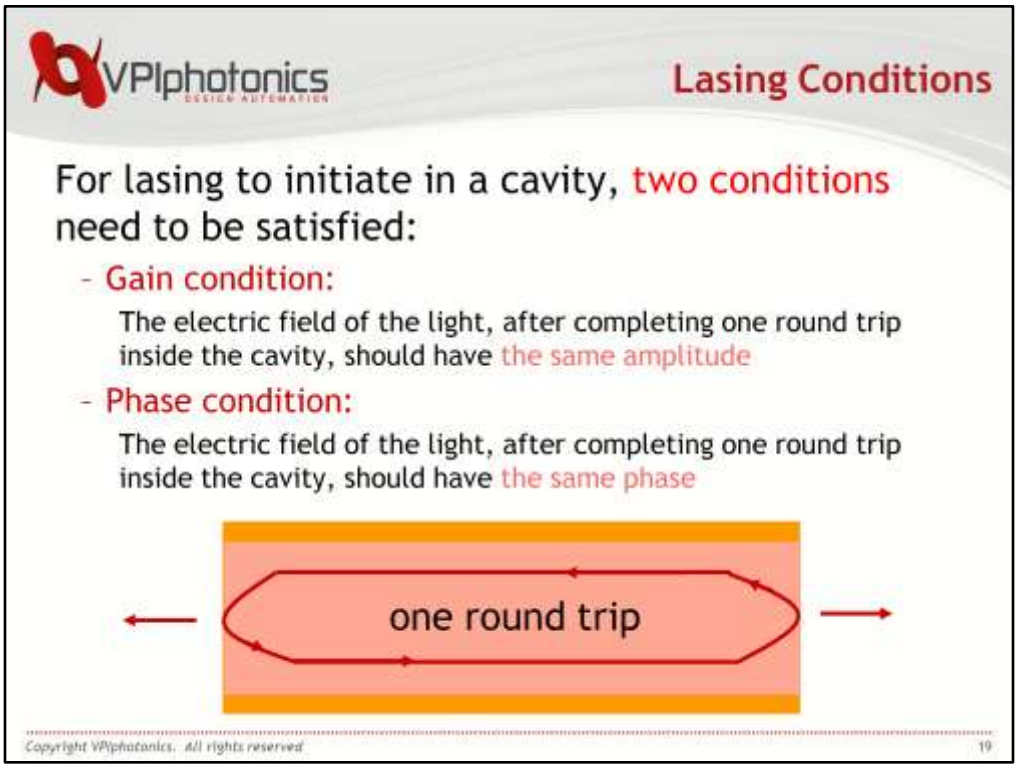


If one mirror is partially transmitting, the transmitted light becomes the output of laser. Stable output is achieved when gain = loss.

Apart from material losses (arising from the absorption of the generated light by the materials used to construct the device and subsequent conversation to heat energy), there are also waveguide losses (due to light lost from not coupling into the guided modes on the laser cavity). Furthermore, any light transmitted through the mirrors also counts as losses. This is important! If the mirror are lossless, then the device would not produce any light output. The losses sustained via the transmission of light from the cavity to the surrounding environment provide the required rate of dissipation of energy required to balance the energy flow into the device.



In earlier slides, it was mentioned that the electrons in ground state (and similar in the excited state) do not occupy exactly the same energy level, but distributed in a band of energy levels centered on a mean level. This means that there is a distribution in

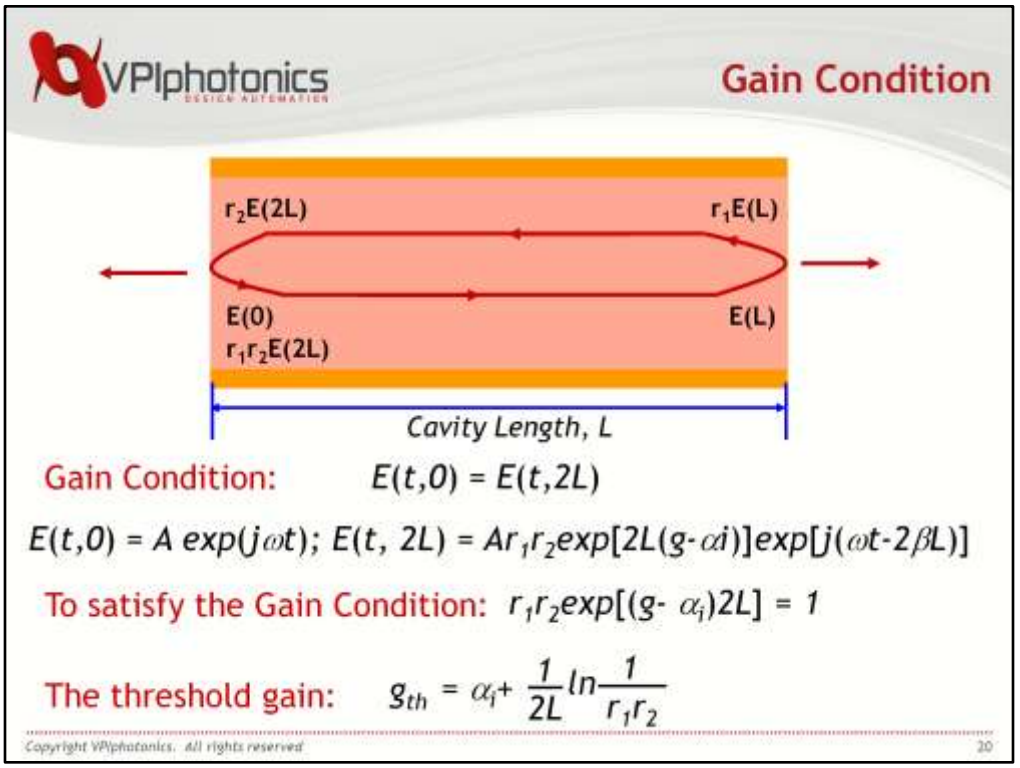


Apart from population inversion and the optical transition processes (absorption, spontaneous and stimulated emission) in the laser material, two more conditions need to be satisfied in order for a laser to produce a coherent output.

These two conditions are determined by the laser cavity, and modifies the spectrum (gain vs frequency curve) of the laser light produced by the optical transition processes. When they are met, the laser is said to “lase” or (or starts “lasing”) and for this reason, these two conditions are termed the “lasing conditions”. They are:

1) Gain condition - the electric field of the light, after completing one round trip inside the cavity (or resonator), should have the same amplitude.

2) Phase condition - the electric field of the light, after completing one round trip inside the cavity, should have the same phase.



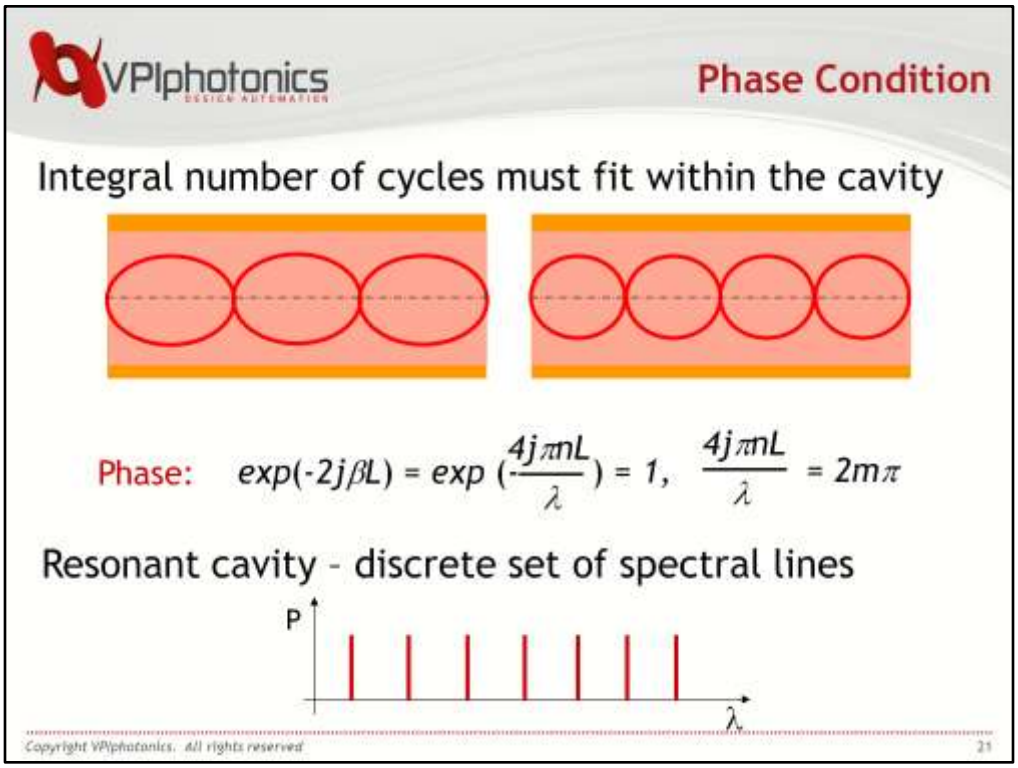
The Gain Condition

This slide shows the mathematical application of the gain condition to deduce certain properties of the laser cavity.

1) the gain condition can be summarised as *E(t,0) = E(t,2L)* i.e. after one round trip (distance *2L* = twice the cavity length), the amplitude must be the same.

2) Substituting for *E(t,0)* and *E(t,2L)* i.e. the travelling wave equation solution, taking into account the reflectivities of the end facets (r_1 and r_2), the effective absorption coefficient of the material in the optical path (of the light) a_i and g is the gain in the active region (in a more detailed derivation, an optical confinement factor Gamma is also included to take into account the proportion of the optical field which lies in the active (gain) region of the laser, but this is neglected here for simplicity at this stage).

3) The gain condition is rewritten in terms of parameters of the laser (

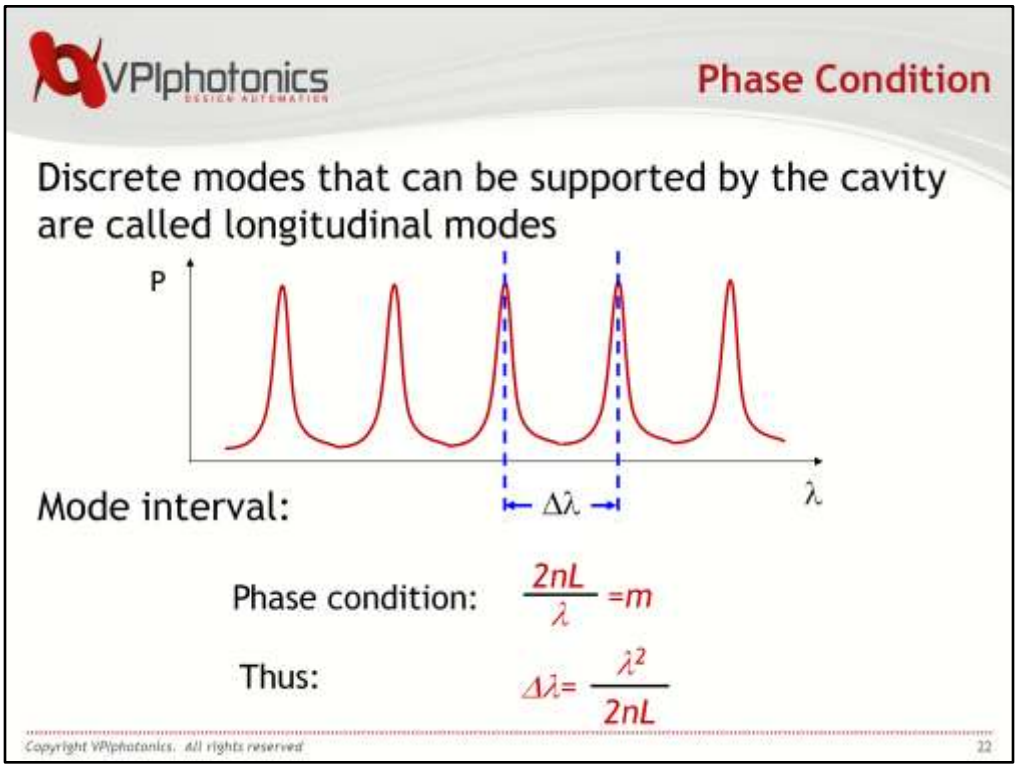


The Phase Condition

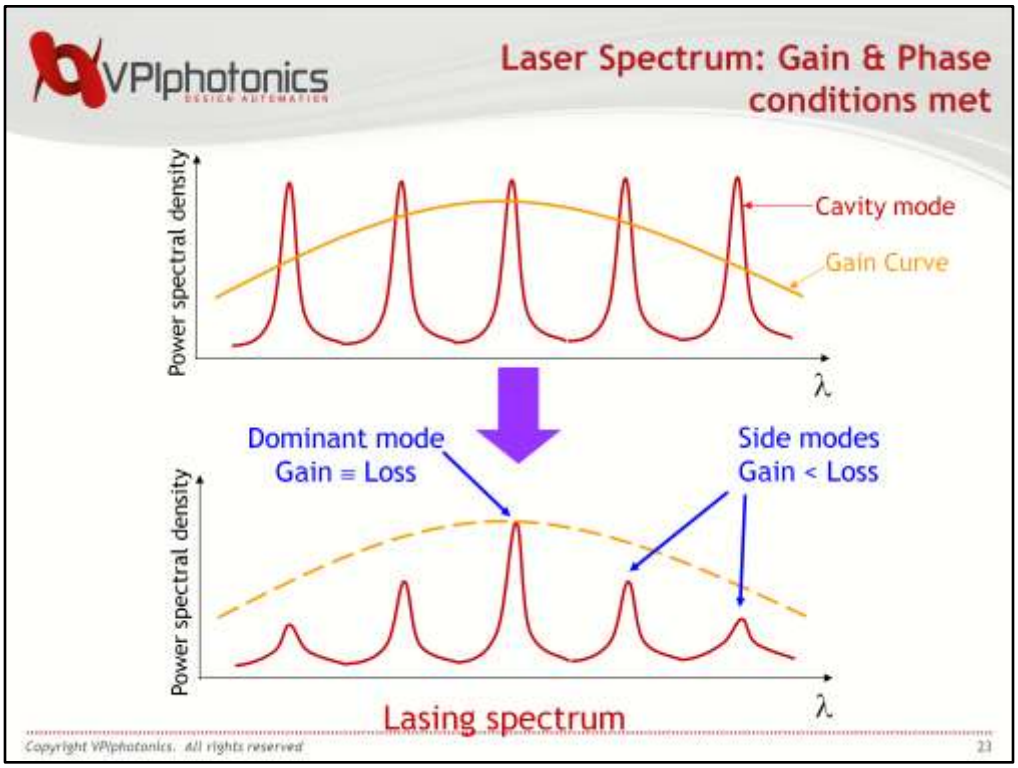
This slide shows the mathematical application of the phase condition to further deduce the properties of the laser cavity.

1) The phase condition states that after one round trip, the optical field must have the same phase. This also means that an integral number of wavelengths must fit within the distance of a round trip (*2L*).
2) Equating the phase part of *E(t,0) = E(t,2L)*, this means that *exp(-2jbL)* = 1 = *exp(2m.pi)*, i.e an integer number of cycles (wavelengths) must fit a round trip of the cavity. Here, β = beta = propagation constant, and *m* is the mode number.
3) Rewriting, obtain: *exp(-2jbL)* = *exp (-4jpi.nL/lambda)* = *exp(2m.pi)*, where *n* = refractive index, *lambda* = wavelength.

Solving for *lambda*, we find that only light of certain discrete (or "resonant") wavelengths are permitted to exist by the cavity – light at other wavelengths are severely attenuated (due to destructive interference).

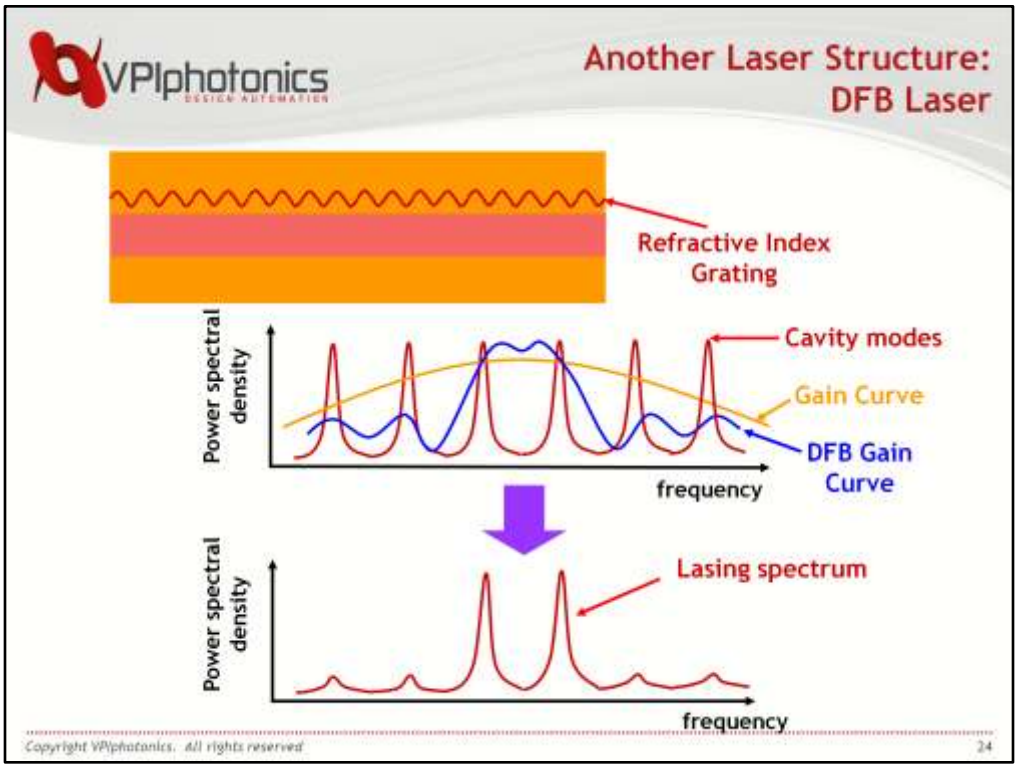


The resonant wavelengths are the discrete modes that are allowed or supported by the laser cavity, and are also known as the “longitudinal modes” of the laser. The spacing between the modes, delta lambda, is given by the expression in the slide and is known as the cavity-mode spacing, or the “free-spectral-range” (FSR) of the cavity.



Meeting the gain condition of the cavity enables the laser to “lase”.
Meeting the phase condition of the cavity results in the laser producing light at discrete (resonant) frequencies. The spacing of these resonant frequencies is determined principally by the length of the cavity (*L*), and the refractive index (*n*) of the material.
These two conditions modify the gain vs wavelength curve (herein referred to as simply the gain curve) obtained from the optical transition processes. Superposing the two set of curves (shown in the top diagram), we obtain the lasing spectrum of the laser (shown in the bottom diagram).
The dominant mode is the cavity (or longitudinal) mode that is closest to the peak of the gain curve, and is sustained as the gain = loss in the laser for that mode. The other longitudinal modes are weaker as gain < loss for them in the laser, and they survive because continual replenishment from spontaneous emission (which still occurs even though stimulated emission is dominant when the laser is lasing).

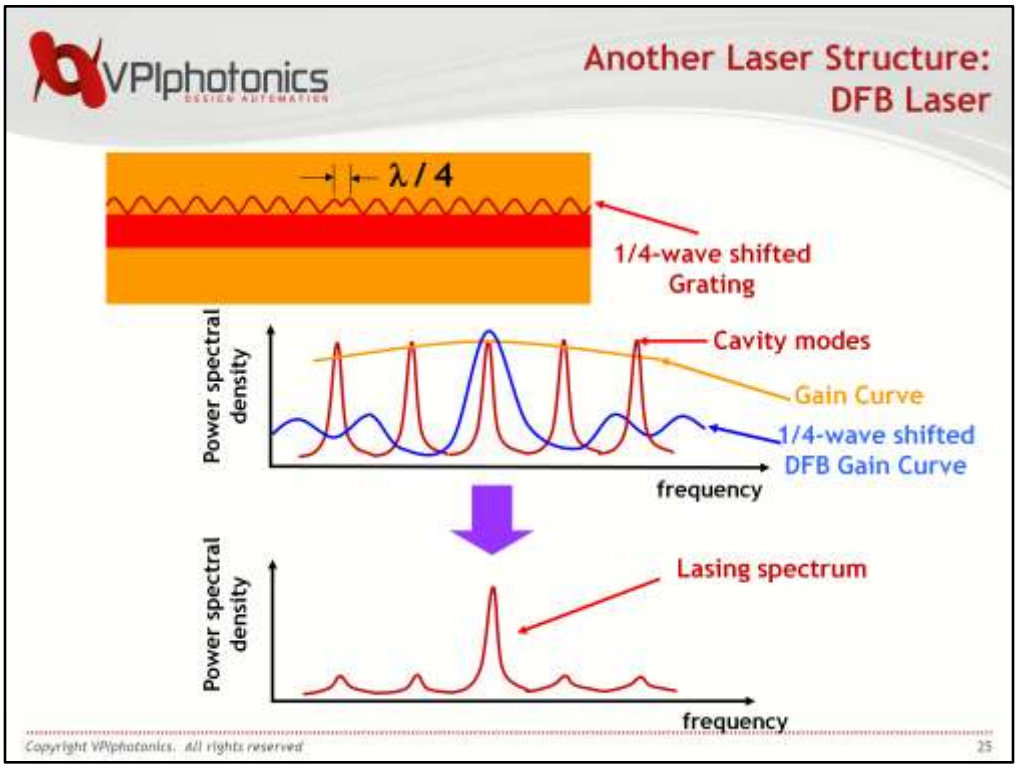
Note that the above spectrum is the typical multi-moded spectrum of Fabry-Perot lasers (lasers made up of active medium sandwiched between two partial mirrors). Other types of lasers are possible (e.g. Distributed Feedback lasers, see next two slides, but these are optional can be left out of the lecture at the instructor’s discretion), but we will not go into the details in this module.



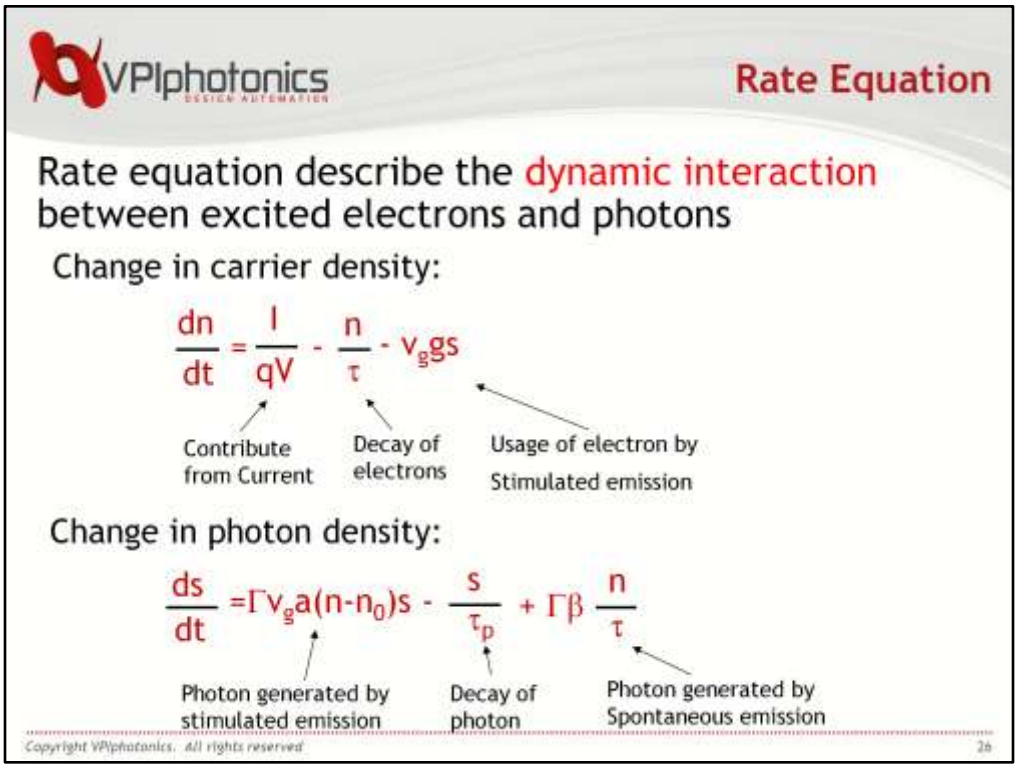
The multimode spectrum of the Fabry-Perot laser is not desirable for long distance and high bit rate systems, due to the effects of intermodal dispersion, which severely restricts the maximum transmission distance and bit-rates of multi-mode laser pulses (signals).
To minimise the effects of dispersion in single-mode optical fiber, a laser diode which provides a single longitudinal mode operation are required for long distance and/or high bit rate systems.
Wavelength selective optical feedback in the laser cavity can produce single longitudinal mode operation, and this is the case with Distributed Feedback (DFB) lasers. Instead of mirrors at either end of the laser cavity, feedback is provided in the form a refractive index grating that coexist with the cavity. The grating (shown diagrammatically as a wavy line above the active region in the top diagram) acts like a “distributed mirror or feedback mechanism”, reflecting photons back into the cavity over a distributed area. Another way to think of this is a series of partially reflecting mirrors spaced evenly along the active medium. Note that the end facets of the DFB laser have to be coated with an anti-reflection coating in order to avoid forming an additional Fabry-Perot type cavity.

Without going into detailed mathematical derivations, the effect of the grating is to produce a DFB gain curve as shown in the middle diagram in the slide. Because of the distributed nature of the reflections (feedback), far fewer cavity modes are allowed by the DFB cavity compared with a Fabry-Perot cavity.

Superposing all three curve gives the lasing spectrum of a DFB laser, which has two dominant modes (bottom diagram). This is an improvement over the multimoded spectrum of a Fabry Perot laser, with greater suppression of the unwanted side modes.

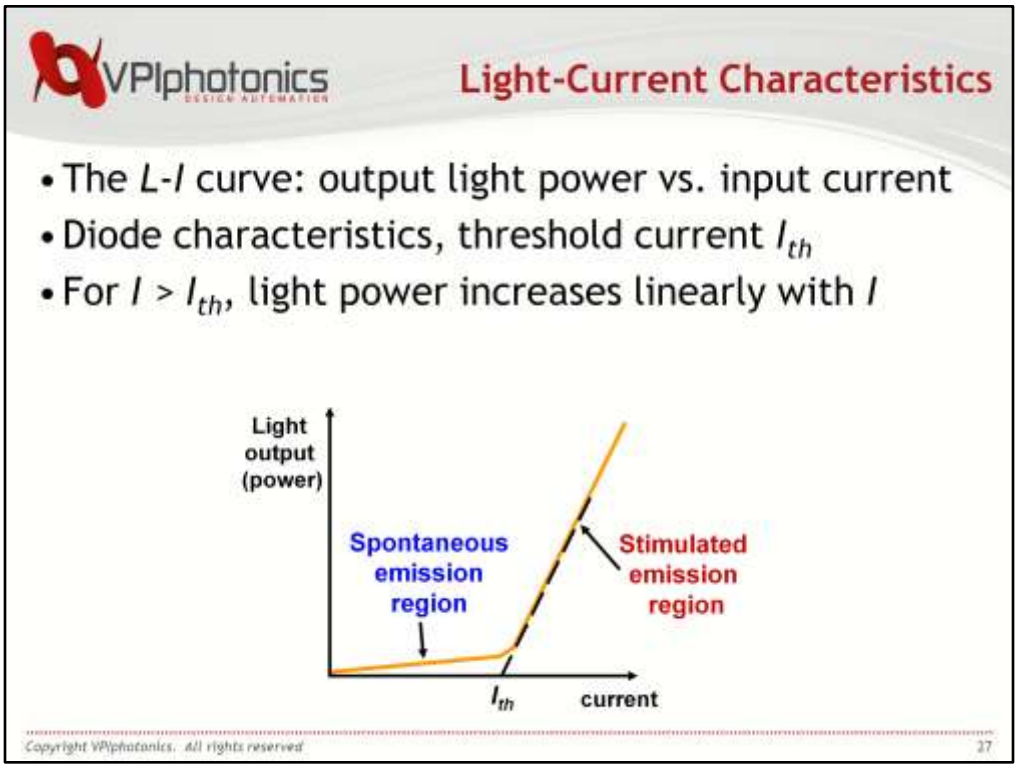


However, the two-moded spectrum of the DFB laser is still unsatisfactory (it will still give rise to intermodal dispersion problems). A trick is to introduce a phase-shift of a quarter of a period (of the grating) in the middle of the DFB cavity. Again, without going into the maths, it can be shown that such a laser (known as a quarter-wave shifted DFB laser) has a lasing spectrum with only a single dominant mode.



The rest of the slides in this module aims to introduce the dynamic effects of lasers without much in-depth discussion of the underlying physics or mathematical derivations. This is necessary so that the students can gain the benefit of the learn-by-simulation module, which includes the dynamic effects of lasers (which are much more interesting than the basic and static behaviour of lasers). There should be follow up modules to present in-depth expositions of the various dynamic effects of lasers.

For this slide, the idea is to show that the dynamic effects of lasers can be modelled by Rate Equations, (which are similar to the Einstein Relations in that the rate equations model the time-dependent relationship between the carriers (electrons) and the photons in the laser). Students are not expected in this one module to grasp the intricacies of the Rate Equation laser model. Rather, the key message is that lasers can be modelled by differential equations that can be solved relatively easily (and dynamic behaviour predicted, as a result).

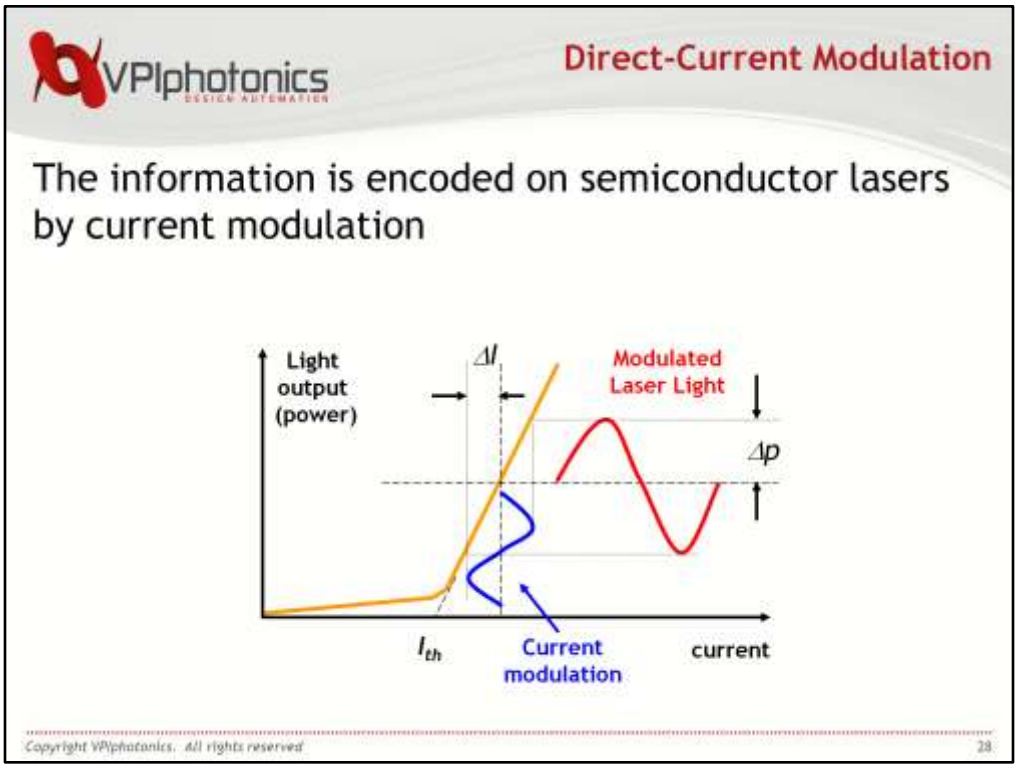


The Light-Current (L-I) curve shows the output (light) power of the laser as a function of the input current into the laser.

As can be seen, this curve is characteristic of a diode (I-V curve, with a threshold bias level V_{th}):

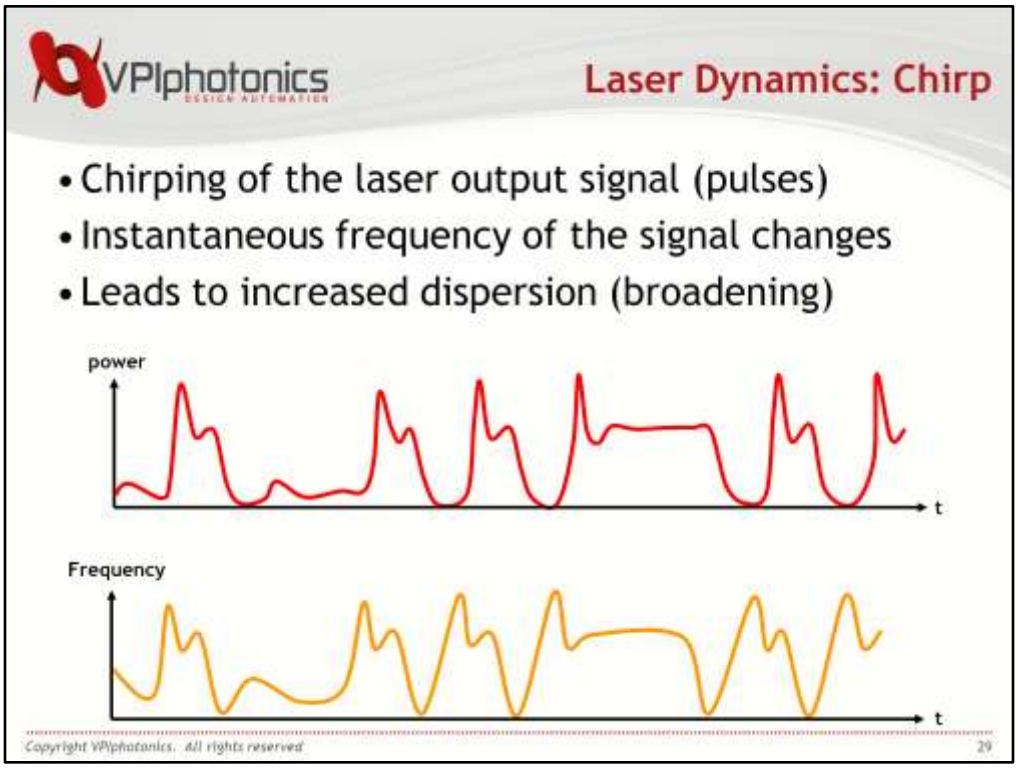
•Below the threshold current, the laser does not produce much output, and spontaneous emission dominates.

•Once the threshold current level is supplied to the laser, it begins to lase, and the output power increases linearly. Here, stimulated emission dominates.



It is obvious from the L-I curve that if the laser is biased in the linear region of the curve, then information can be modulated onto the laser output simply by modulating the input current. This is known as the direct current modulation (or simply “direct modulation”) of a laser.

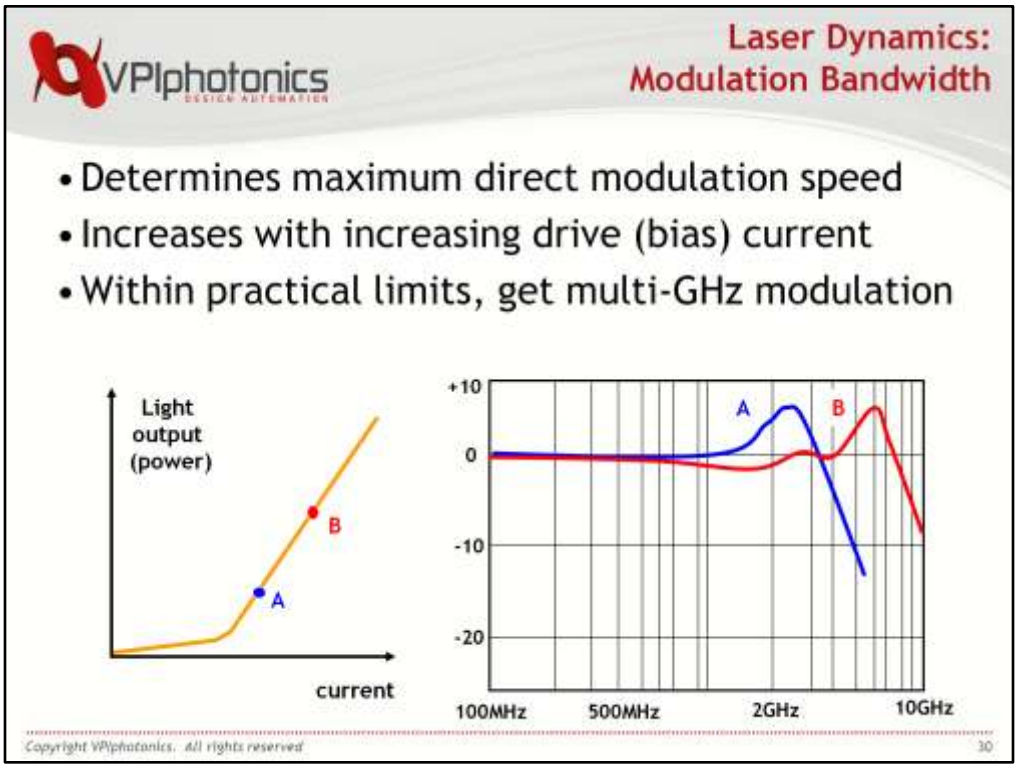
However, there is a limit to how fast the laser can be modulated this way, and this depends on the modulation bandwidth of the laser (see a later slide). Also, the light pulses generated by direct modulation suffers from “chirp” (see next slide)



Laser chirp is the change in the instantaneous frequency of the laser signal. The top diagram shows the output light signal from a directly modulated laser. The bottom diagram shows the instantaneous frequency of the same signal, which is changing (and somewhat resembles, but not exactly, the shape of the signal itself). This is caused by the fact that the refractive index (R.I.) of the active medium of the laser actually depends on the amount of carriers (electrons) present. As the current is modulated, the number of carriers changes with the modulation, which changes the R.I.

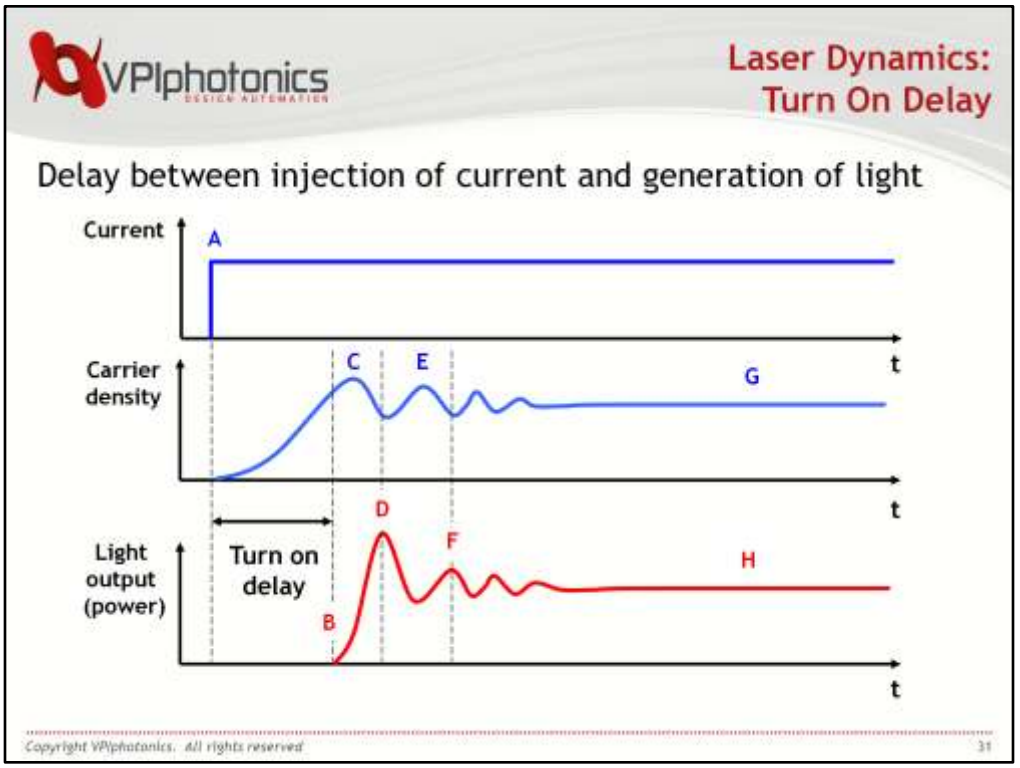
If there is no chirp, the instantaneous frequency of the signal should be constant (= the frequency of the dominant mode of the laser).

Chirp is undesirable in fiber-optic transmission systems as chirped light pulses suffer greater from the effect of fiber dispersion and this can severely reduce the maximum achievable transmission distance and bit-rate of the system. As such, for high data rates and long distances, direct modulation of lasers is not suitable.



The modulation bandwidth of a laser determines how fast it can be directly modulated. Research and experiments have shown that this can be increased by driving the laser at higher currents (point B and curve B in the diagrams above). There are practical limits, of course – aside from current and heat dissipation limits of the device, too high a photon flux (i.e. too much light) from the laser actually destroy the end-facets of the laser (thereby ruining the device). This is due to the localised heating at flaws in the end-facets.

However, with improved design, it is possible to make lasers that are capable of multi-GHz direct modulation speeds.



When the current is injected into the laser, light is not instantaneously produced. A short but finite period of time elapses before the laser switches on and starts producing light. This time period is known as the “turn on delay” of the laser.

An explanation of what happens (if there is enough time in the lecture to go through this for the students):

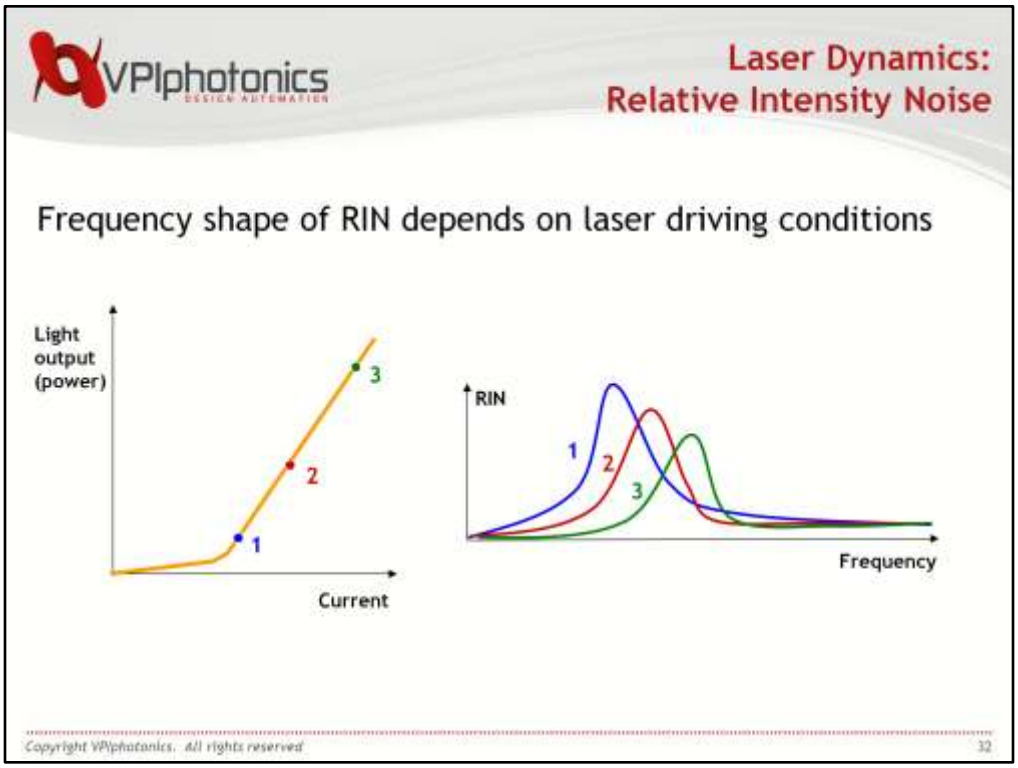
At point A (in the top graph): Current is injected into the laser. The carrier density (middle graph) in the laser starts to build up.

At point B (in the bottom graph): The carrier density is sufficiently high and population inversion is reached. Stimulated emission begin occur and light is produced. Light power (which is proportional to the photon density) starts to increase, and the build up of carrier density slows down eventually peaking at point C (middle graph).

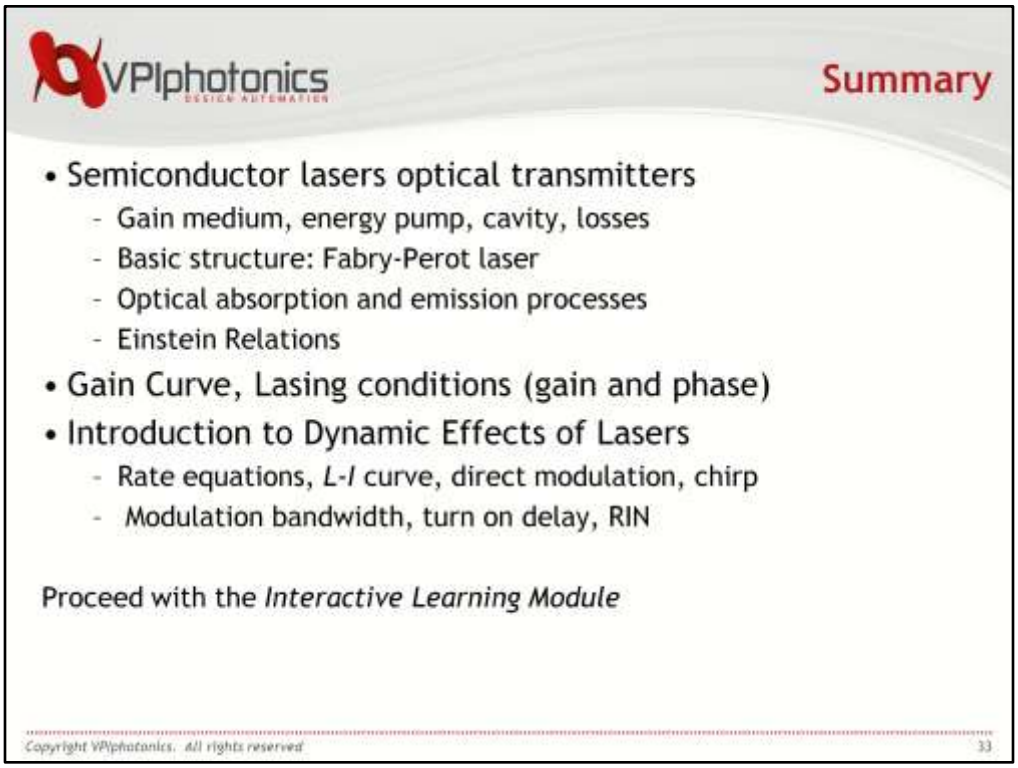
As the light power continues to increase, the carrier density starts to decrease again. This is due to the rate of stimulated emission outstripping the rate of carrier injection into the laser.

Eventually, the light power peaks at point D (bottom graph) and even starts to decrease again. The decrease in light power is due to the shortage of carriers to sustain the rate of stimulated emission.

Since the rate of stimulated emission is now reduced (below the rate of carrier injection into the laser), the carrier density builds up again, peaking at point E. With sufficient carriers, the rate of stimulated emission increases again, and the light power increases again, peaking at point F. Note that the peaks E and F is lower



The power emitted by a semiconductor laser fluctuates around its steady state value. The spectrum of these fluctuations can be measured by detecting the laser output using a wide-bandwidth photodiode and spectrum analyzer.
The intensity noise at a given frequency is characterized by the relative intensity noise (RIN), which may be defined in terms of the mean square power fluctuation and the mean optical power squared.



Summary of the key ideas of the lecture part of this module